

SE LAB1

Problem Statement #52 — Community Tool & Equipment Library

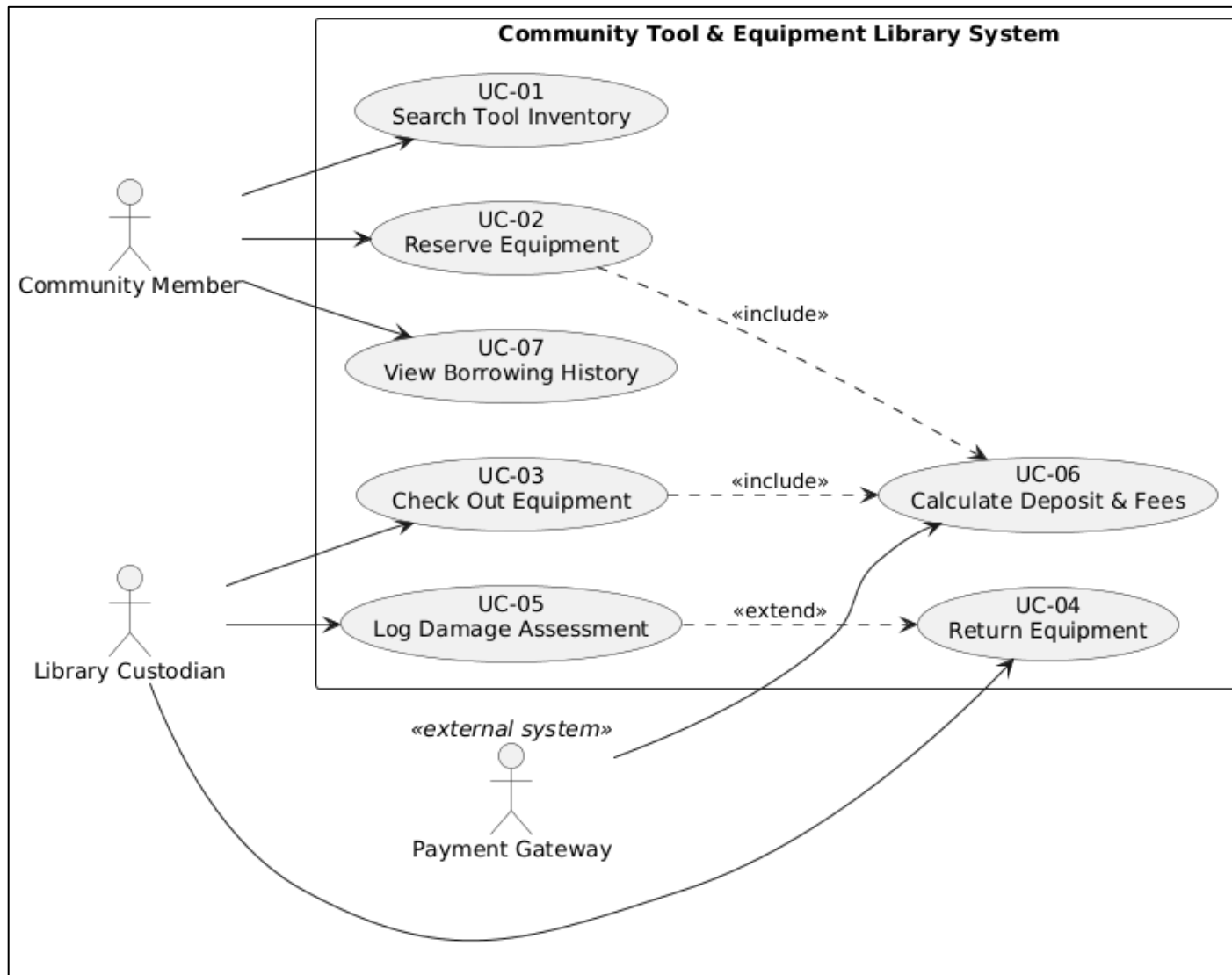
1. Complete Requirements Table

2. Actors: Community Member, Library Custodian, Payment Gateway (external system)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate required security deposits based on tool category, enforce loan duration limits, and calculate late return fees.	High	Pass: Tool reservation confirmed with deposit hold. Fail: Member with overdue tools borrows additional high-value equipment.	Protects shared assets and ensures financial accountability before high-value equipment leaves a depot.
FR-002	Functional	The system shall allow a Community Member to reserve an available tool for a specified checkout window, holding it for pickup until a defined grace-period deadline.	High	Pass: Reserved item is hidden from other members' search results until the grace period expires. Fail: Two members can reserve the same physical unit at the same time.	Prevents double-booking of shared physical inventory across neighborhood depots.
FR-003	Functional	The system shall allow the Library Custodian to process an equipment return, record the return timestamp, and log a damage assessment (condition rating and notes) against the loan.	High	Pass: Return is timestamped and any damage note is attached to the loan record. Fail: A returned item is marked "available" while an open damage assessment is unresolved.	Creates an auditable condition history so deposit deductions and future loan decisions are defensible.
FR-004	Functional	The system shall let Community Members search and filter the inventory catalog by tool category, depot location, and availability date.	Medium	Pass: Results show only items physically present and unreserved at the chosen depot/date. Fail: A reserved or checked-out item appears as available.	Supports the real-time availability goal (NFR-001) and avoids wasted trips to a depot.
FR-005	Functional	The system shall maintain a per-member borrowing history, including active loans, past loans, and any outstanding deposits or fees.	Medium	Pass: Custodian sees a member's full loan/fee history before approving a new checkout. Fail: A member with an overdue fee is approved for checkout without the balance being shown.	Gives custodians the context needed to enforce the overdue-block rule in FR-001.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Non-Functional	The tool inventory search must display real-time physical availability across neighborhood depot lockers, meeting target latency and security standards.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Members need trustworthy, current availability data before traveling to a depot; stale data undermines the sharing model.
NFR-002	Non-Functional	Damage assessment records and deposit/fee transactions shall be stored as immutable, timestamped audit entries retained for a minimum of 3 years.	High	Pass: Editing a submitted damage record creates a new versioned entry rather than overwriting the original. Fail: A damage note or fee amount can be silently altered after submission.	Deposit deductions can be disputed by members; an immutable audit trail resolves disagreements fairly and limits library liability.

2. UML USE CASE DIAGRAM



3. Use-Case Flow Specification

Use Case: UC-02 Reserve Equipment

Primary Actor: Community Member | Secondary Actors: Payment Gateway (via included use case)

Includes: UC-06 Calculate Deposit & Fees

Preconditions

- The Community Member has an authenticated, active account.
- The member has no overdue equipment on their account.
- At least one unit of the requested tool is available and unreserved at a depot.

Postconditions (Success)

- The reserved equipment's status changes to "Reserved" and is hidden from other members.
- A security deposit hold matching the tool category is placed via the Payment Gateway.
- A reservation confirmation with a pickup deadline is sent to the member.

Main Success Scenario

1. Member searches the inventory and selects an available tool.
2. Member selects a reservation window and a depot locker for pickup.
3. System checks the member's account for overdue equipment.
4. System calculates the required security deposit and applicable loan duration limit (includes UC-06 Calculate Deposit & Fees).
5. System sends a deposit-hold request to the Payment Gateway.
6. Payment Gateway authorizes the hold and returns a confirmation code.
7. System reserves the equipment and marks it unavailable to other members.
8. System generates a reservation confirmation with the pickup deadline and sends it to the member.
9. Use case ends successfully.

Alternate Flow A — Member Has Overdue Equipment (triggered at step 3)

- a. System detects an outstanding overdue loan on the member's account.
- b. System rejects the reservation request and displays the overdue item(s) and any fees owed.
- c. Member is prompted to return or settle the overdue item before reserving new equipment.
- d. Use case ends without a reservation being created.

Alternate Flow B — Deposit Hold Declined (triggered at step 6)

- e. Payment Gateway declines the deposit-hold request.
- f. System displays a "payment method declined" message to the member.
- g. Member is prompted to try another payment method; after 2 failed attempts the reservation is cancelled and the member is notified.